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Week 2 Assignment

How to Run

In the lab2 directory, you can run "go run ." to run the Search.go . The Search.go has part 1 and 2 of the assignment where it measures the sequential word count and the go routines word counting.

For part 4, where it uses the RPC library. You have to "cd server", then in the server folder, you can "go run ." to run the server. Open a new terminal, then navigate into the client folder. Inside the client folder, you can run "go run .", to run the client code.

Table

Part	Timer
1	12.124875ms.
2	10 workers. It took 2.043792ms.
3	rpc: 43478. It took 19.321292ms rpc and 8 go routines: 43478. It took 4.714041ms.

Design Decisions

- search_functions handles all search logic
- main handles input, file reading, timing, and output
- The number of workers is configurable and defaults to at least one for safety.
- Each chunk includes an overlap of len(term)-1 characters so matches that cross chunk boundaries are not missed.
- Goroutines communicate results through a channel, and a WaitGroup ensures all workers finish before results are combined.
- The main design challenge is preventing double counting in overlapping regions; each match should belong to only one worker.